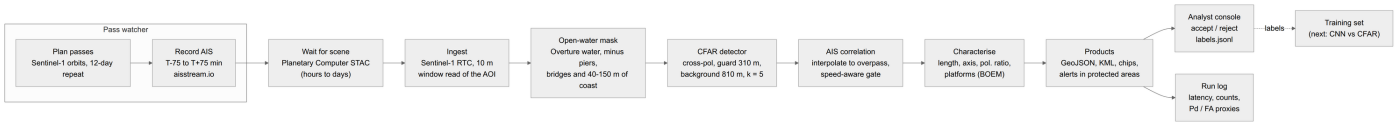
# Ships in Sentinel-1 radar, checked against AIS

## Operational-style ship detection and AIS correlation for maritime domain awareness

A service built to run by itself after each Sentinel-1 pass: it detects ships, decides which ones AIS accounts for, and flags the rest for an analyst. It is scored as an operator would use it: probability of detection (Pd) at a fixed false-alarm rate. The validation uses **12** Sentinel-1 passes from 2024 over US waters, where a full year of AIS is public (NOAA MarineCadastre), plus a locked set of **36** passes in three areas never used to choose a production detector setting. Production uses classical **constant false-alarm rate (CFAR)** detection; a learned two-stage detector finds more ships at strict false-alarm budgets on average (Pd **0.32** ± 0.08 vs **0.22** at 1 FA; the margin varies by seed) and ties at 5 FA and at production's false-alarm rate, small craft included, but failed the promotion gate. A second one (v6, a Sentinel-1 ResNet-50) passes the gate, but a pre-registered test in three new areas found it worse than CFAR at equal recall; the gate gained an equal-recall rule, which v6 still passes on the original test passes. Built by Shivaji Patil in October and November 2026; an AI coding agent wrote much of the code under his direction. Every result comes from scripts in a private code repository (the Danish study's from a separate one); the learned-detector results also need trained weights not included in git.

## How it is built to run

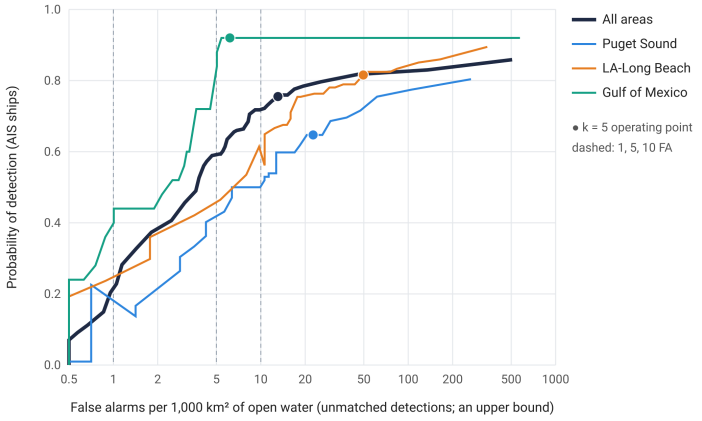



Timers record live AIS around each Sentinel-1 pass of Vancouver and St. John's. When the scene is published, it is masked to open water and searched with CFAR. Targets are correlated with AIS, characterised and turned into products and alerts, then reviewed in the analyst console.

## Validation: 12 passes in 2024, NOAA AIS as truth

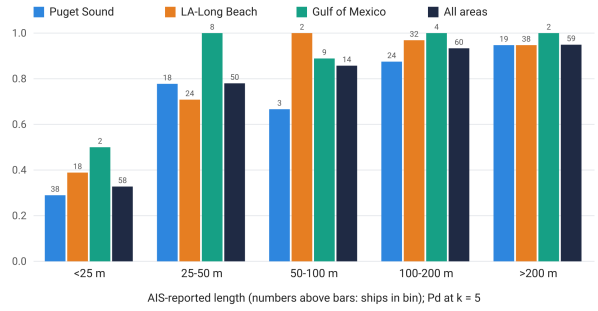
| Area | Passes | AIS ships | Water km² | Pd at 1 / 5 / 10 FA | Pd, k=5 | FA, k=5 | Unmatched | Platforms |
|---|---|---|---|---|---|---|---|---|
| **All areas** | 12 | 226 | 10,469 | **0.22** / **0.63** / **0.77** | **0.81** | **13.1** | 137 | 120 |
| Puget Sound | 4 | 92 | 1,412 | 0.25 / 0.45 / 0.55 | 0.72 | 22.7 | 32 | 0 |
| LA–Long Beach | 4 | 109 | 1,125 | 0.25 / 0.44 / 0.64 | 0.84 | 50.7 | 57 | 0 |
| Gulf of Mexico | 4 | 25 | 7,933 | 0.36 / 0.92 / 0.96 | 0.96 | 6.1 | 48 | 120 |

AIS ships on open water at least 150 m from the mapped coast. FA: detections that no AIS ship accounts for, per 1,000 km² of open water. An upper bound: vessels without AIS count against it. Platforms: detections within 150 m of a standing BOEM platform, not scored. A match must lie within 300 m of a region stretched along track by a moving ship's expected image shift; random detections would reach Pd **0.04**.




Operating curves. The detector is re-run at each threshold, so every point is a real fixed-threshold detector. Dots mark k = 5.

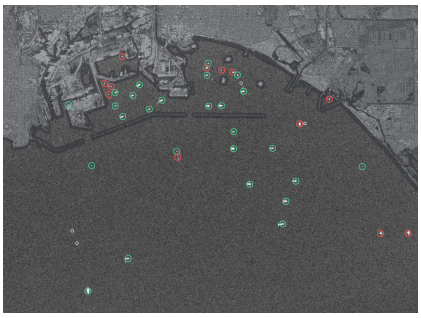
- **Position error:** stationary ships median **42 m**, p90 **139 m**. Moving ships median **152 m**, p90 634 m: a moving ship is imaged shifted along the track by about 115 s times its radial speed. On the locked set, CEP90 for stationary ships is 63 m, and a 90% position ellipse on every detection holds **90%** of AIS positions.



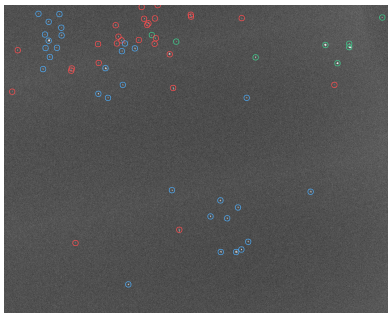

Pd at k = 5 by AIS length, all areas: **0.36** under 25 m, **0.83** at 25–50 m and **0.95**–**0.98** over 100 m.

- **Locked test** (Massachusetts Bay, Chesapeake entrance, New York Lower Bay; 36 passes, 284 ships): Pd **0.69** at k = 5, **0.94** for ships of 50 m and more, **13.2** FA. Only Massachusetts Bay (Boston) is unseen water; the other two overlap development areas. Its passes were later among the 129 used to fit the second fresh test's candidate thresholds (a false-alarm budget), which were rejected. The list was fixed in local commit 2f818e9 (24 Oct 2026) and first pushed to GitHub on 3 Nov with its scoring, so its order rests on the commit dates. Its first scoring exposed a truth fault (AIS clipped to the box), since fixed for every set; the detector did not change.

## QUALITY CONTROL AND WHAT IT CHANGED



LA–Long Beach, 20 Apr 2024: **green** matched to AIS, **red** unmatched; white rings are AIS ships not detected.

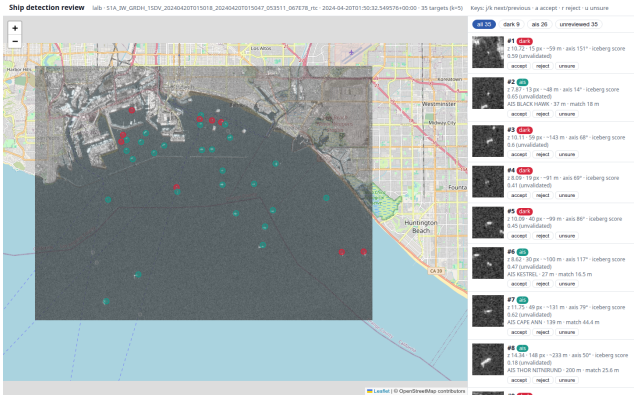


Gulf of Mexico platforms, 12 Oct 2024: **blue** within 150 m of a BOEM platform (30 in this pass), kept out of false alarms.

- **Coastal buffer.** At Harbour Twin's 40 m buffer, Puget Sound's unmatched targets lined the shore in the QC images: private docks and shore returns not on the map (**350** unmatched at 40 m, **32** at 150 m). The cost: ships within 150 m of the coast are not scored (272 AIS ships at 40 m, 226 at 150 m). Re-chosen on the 23 training passes, the rule picks 300 m: on the test passes Pd **0.66** at 5 FA (150 m: **0.63**), but only 201 of 226 ships scored. So 150 m is not flattered; Pd alone cannot pick the buffer.

## CHARACTERISATION AND ANALYST REVIEW

- **Length and axis** come from the second moments of each target's pixels. Against AIS for **182** matched ships, radar length runs long, with a median bias of +9 m and a median error of 16 m (correlation **0.84**). The axis is within 20° of the AIS heading for **77%** of ships; the median error is 4°.
- **Platforms:** **120** Gulf detections fall at standing BOEM structures and would otherwise count as dark vessels.
- **Iceberg vs ship (St. John's):** a rule-based score from polarisation ratio, compactness and size. It is **not validated**: no official sightings with clear terms overlap these scenes.
- **Products per pass:** GeoJSON and KML, 640 m image chips, alerts for unmatched targets inside marine protected areas (CPCAD), and a run log with latency. Each pass is also written in the Harbour Twin viewer's format.




Analyst console on 127.0.0.1: map, target list with chips, accept or reject. Decisions are saved as labels. Base map © OpenStreetMap contributors.

## SECOND STUDY: DANISH WATERS AT SCALE

- **Physics-based matching:** on 2,585 fresh AIS matches, the along-track offset follows the Doppler azimuth shift (slopes **0.95** / **0.94**, theory 1.0); correcting it cuts the median offset from about 200 m to 60–65 m.
- **VH beats VV:** over 70 frames, the unmatched list halves (27.1 to 13.6 per 1,000 km$^2$) at similar recall.
- **Generalisation:** recall 81% in the straits, 92% in the North Sea, and 72% on 94 frames from a held-out month (May).
- **Foundation models:** on 2,739 AIS-labelled chips, a fine-tuned Sentinel-1 SoftCon backbone gives vessel-type macro-F1 **0.52** ± 0.04 (ImageNet **0.48** ± 0.03, within one SD) and learned length error 19–27 m (CFAR 44 m).

## OPERATIONS, LIMITS AND NEXT STEPS

- **Live:** built to run by itself after each pass; first live pass (Vancouver 22 Oct; restarted 5 Nov): 46 targets, 27 of 33 AIS ships found, 52 s to process, 12 h after sensing (Planetary Computer's publishing delay; near-real-time needs Copernicus NRT). Changes reach the service only through an MLflow registry and promotion gate (one recorded override, v2), and each pass records the detector version.
- **Limits:** AIS is incomplete truth; boats under 25 m are often missed at 10 m; ships within 150 m of the coast are not scored; the US AIS-scored passes are all from 2024.

1. **Detector:** a false-alarm budget did not transfer; a structure map from each region's own history cut dark-list false alarms 34% in five new areas (41% as first scored) and 65% in five structure-rich ones, mooring spots unmasked, but only 9% in five dense, already-listed fields (no demonstrated improvement); an alongside-vessel flag failed its false-alarm bar and, recalibrated, both. Grand Banks, no AIS: 3 of 4 installations found (bar: all four), plus West White Rose. Norway/UK, 300 m radius: 15 of 16 swinging FPSOs vs 12 (bar: +5).
2. **Next:** analyst review of map spots; icebergs; RCM.

Data: contains modified Copernicus Sentinel data (2024–2026). Sentinel-1 RTC from Microsoft Planetary Computer (processed by Catalyst), licensed by Microsoft under CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/); images here are modified. AIS: Office for Coastal Management (2026), Nationwide Automatic Identification System 2024, NOAA InPort item 73064, from U.S. Coast Guard Navigation Center data (U.S. Government data). Live AIS: aisstream.io. Water and piers: Overture Maps (incl. OpenStreetMap data, ODbL). Platforms: BOEM, C-NLOER, Government of NL; Sodir FactPages (NLOD 2.0); contains information provided by the North Sea Transition Authority and/or other third parties. Protected areas: CPCAD (ECCC), Open Government Licence – Canada. Danish study: Danish Maritime Authority AIS. SoftCon weights: CC BY 4.0. Related: Harbour Twin.